

**SECTION – A: MULTIPLE CHOICE QUESTIONS - CHOOSE ONLY ONE CORRECT AND BEST ANSWER FOR EACH QUESTION AND CIRCLE THE LETTER IN FRONT OF IT. ANSWER ALL QUESTIONS (30x1 = 30 MARKS)**

1. Which one of the following is mostly responsible for the maintenance of protein secondary structure?
  - a. Hydrophobic interactions
  - b. van der Waals forces
  - ☒ c. Hydrogen bonds
  - d. Disulfide cross-links
  - e. Salt bridges
2. Which one of these best describes all amino acids that are used to synthesize human proteins (with exception of glycine). They have.....
  - a. An aromatic group
  - b. A hydroxyl group
  - c. An asymmetric carbon in the D-configuration
  - ☒ d. An asymmetric carbon in the L-configuration
  - e. An asymmetric  $\alpha$ -carbon
3. .... is the type of fibrous protein found in hair, nails and outer epidermal layer of mammals.
  - a. Elastin
  - ☒ b. Keratin
  - c. Collagen
  - d. Procollagen
  - e. Proelastin
4. In trying to check whether a patient had sickle anemia, the doctor requested the laboratory to determine the electrophoretic spectrum of the proteins in the patient's blood. What physical/chemical properties of proteins is this method based on?
  - a. Optical activity
  - ☒ b. Solubility
  - c. Presence of charge
  - d. Viscosity
  - e. Ability to denaturation
5. The catabolism of hemoglobin.....
  - a. results in the formation of protoporphyrinogen
  - b. results in the formation carbon dioxide
  - c. occurs in red blood cells
  - d. is the only source of bilirubin
  - ☒ e. involves the destruction of the porphyrin ring

6. Which of the following statements is correct about most enzymes?
- a. increase the activation energy
  - ☒ b. can make a reaction happen faster
  - c. are used once and need to be replaced after catalyzing a reaction
  - d. they change the position of the reaction's equilibrium
  - e. are not specific to the type of substrate they receive
7. Which of the following statements about sickle cell anemia is correct? It is.....
- ☒ a. an autosomal recessive disorder
  - b. an autosomal dominant disorder
  - c. a sex linked disorder
  - d. due to increased solubility of Hemoglobin
  - e. due to reduced deformability of red blood cells
8. Which one of the following statements, concerning asparagine is correct?
- ☒ a. Contains an amide group
  - b. Has Asp as its three-letter abbreviation
  - c. Classified as an acidic amino acid
  - d. Has a charged polar side chain
  - e. Doesn't have a chiral carbon
9. Which amino acid is capable of forming disulfide bonds?
- a. Glycine
  - b. Proline
  - c. Histidine
  - ☒ d. Cysteine
  - e. Isoleucine
10. Which of these groups of amino acids are entirely hydrophobic?
- ☒ a. Phenylalanine, Valine, Isoleucine
  - b. Histidine, Glycine, Valine
  - c. Proline, Aspartic acid, Arginine
  - d. Tryptophan, Glutamine, Alanine
  - e. Phenylalanine, Serine, Tyrosine
11. ....is associated with a condition known as phenylketonuria.
- a. Proline
  - b. Methionine
  - ☒ c. Phenylalanine
  - d. Histidine
  - e. Valine

12. A..... amino acid yields acetoacetyl CoA upon catabolism of its carbon skeleton
- a. Glucogenic
  - ☒ b. Ketogenic
  - c. Hydrophobic
  - d. Polar
  - e. Polar and hydrophobic
13. ....is an amino acid the contains an imidazole aromatic ring
- a. Tyrosine
  - b. Phenylalanine
  - ☒ c. Histidine
  - d. Proline
  - e. Tryptophan
14. .... exist in two non-superimposable mirror images of each other
- a. Anomers
  - b. Epimers
  - ☒ c. Enantiomers
  - d. Mutaroters
  - e. Chiral carbons
15. A chiral carbon of an amino acid arises from the fact that it.....
- a. is symmetric
  - b. is bonded to both amino and carboxylic groups
  - ☒ c. is bonded to four different chemical groups
  - d. has net charge of zero
  - e. assumes an L configuration
16. Which ONE of the following amino acid substitutions most likely causes a change in a protein's tertiary structure?
- a. Isoleucine to Arginine
  - ☒ b. Arginine to Lysine
  - c. Valine to Leucine
  - d. Threonine to Serine
  - e. Asparagine to Glutamine
17. In a polypeptide, peptide bonds are formed between.....
- a. two carboxyl groups
  - b. two aromatic groups
  - ☒ c. a carboxyl group and an amino group
  - d. an ester and an amine group
  - e. two amino groups

18. Most of the amino acids found in humans are.....

- ☒ a. L-isomers
- b. D-isomers
- c. Optical isomers
- d. D and L- isomers
- e. Optically inactive

19. ....primarily determines the function of a protein

- a. Type of amino acids
- ☒ b. Spatial conformation
- c. Molecular weight
- d. Chirality of the carbon atoms
- e. Affinity for water

20. Which one of the following amino acids is found in the  $\beta$ -bends as kinks?

- a. Glycine
- b. Alanine
- c. Glutamic acid
- ☒ d. Proline
- e. Valine

21. What type of reactions are catalyzed by hydrolases?

- a. Addition to double bonds
- b. Formation of bonds with ATP cleavage
- c. Transposition of functional groups between substrates
- ☒ d. Transfer of hydrogen atoms between substrates
- e. Hydrolytic cleavage of substrates

22. One of these factors can influence the charge of an amino acid group in the active site of an enzyme

- a. The presence of a coenzyme
- b. The presence of a product
- ☒ c. pH of the medium
- d. Pressure
- e. Temperature

23. Which of the following is not correct about heme?

- ☒ a. Protoporphyrin is bound to a single iron atom in its  $\text{Fe}^{3+}$  state
- b. Iron atom has 6 coordination bonds
- c. Heme consists of a complex organic ring structure, protoporphyrin
- d. Heme is found in a number of oxygen transporting proteins
- e. One of the positions is coordinated to the side chain of a histidine residue

24. Myoglobin is mostly abundant in the.....

- a. Skin
- b. Bones
- ☒ c. Muscles
- d. Nerves
- e. White blood cells

25. Which of these statements is correct about the role of heme in cytochromes?

- a. Functions as part of the active site
- b. Binds oxygen irreversibly
- ☒ c. Functions as an electron carrier
- d. Functions as a competitive inhibitor in reactions
- e. Functions as an activator

26. When the temperature of an incubating environment is raised from  $0^{\circ}\text{C}$  up to  $40^{\circ}\text{C}$  the activity of the human enzyme is usually increased. Which one of the following explains the cause?

- a. The probability of the Enzyme-substrate formation is increased
- ☒ b. The probability of the Enzyme-product formation is increased
- c. The denaturation of the enzyme is reduced
- d. The substrate molecule charge is changed
- e. The enzyme action specificity increases

27. Essential amino acids are.....

- a. synthesized by the body
- ☒ b. obtained from the diet
- c. not involved in the formation of peptides
- d. optically inactive
- e. all acidic

28. In the regulation of enzyme activity, which one of the following is the basis of feedback inhibition?

- a. Covalent modification
- ☒ b. Allosteric regulation
- c. Full proteolysis
- d. Partial proteolysis
- e. Ionic modification

29. Amino acids that make up portions of transmembrane proteins are most likely to be.....

- a. Essential
- b. Non-essential
- ☒ c. Hydrophobic
- d. glycosylated
- e. acetylated

30. Patients homozygous for hemoglobin C have a condition that arises from.....
- a. substitution of lysine by glutamic acid in the  $\beta$ -chain
  - ☒ b. substitution of glutamic acid by lysine in the  $\beta$ -chain
  - c. substitution of valine by glutamic acid in the  $\beta$ -chain
  - d. substitution of a polar amino acid by a non-polar one in the  $\beta$ -chain
  - e. substitution of one of the  $\alpha$ -chains by a  $\beta$ -chain

**SECTION B: DESCRIPTIVE ANSWER TYPE/SCENARIO BASED QUESTIONS.**  
**ANSWER ALL QUESTIONS. FOR EACH ANSWER, ILLUSTRATIONS WITH**  
**DIAGRAMS (WHERE POSSIBLE) WILL ATTRACT MAXIMUM MARKS ( TOTAL**  
**MARKS=50 MARKS)**

1. A 50-year-old male patient was brought to an emergency department of Ndola Teaching Hospital with the diagnosis of myocardial infarction, following determination of different forms of Lactate dehydrogenase (LDH) iso-enzymes.

i. What are iso-enzymes? (0.5 mark)

...They are different forms of the same enzyme catalyzing the same reaction...

ii. Are iso-enzymes organ specific? (0.5 mark)

...yes...

iii. Do they vary in distribution from one organ to the other? (0.5 mark)

...yes...

iv. How is iso-enzyme determination useful in differential diagnosis? (2 marks)

The elevation of an iso enzyme in plasma gives a differential diagnosis of malfunction of that organ e.g. LDH, for myocardial infarction...

v. How many iso-enzymes make up Lactate Dehydrogenase (LDH)? (1 mark)

5 iso enzymes make up lactate dehydrogenase LDH<sub>1</sub>, LDH<sub>2</sub>, LDH<sub>3</sub>, LDH<sub>4</sub>, LDH<sub>5</sub>...

vi. Why is measurement of an iso-enzyme activity, as compared to measurement of total enzyme activity, preferred in most cases? (2 marks)

...iso enzyme activity is organ specific...

vii. Name a technique used to separate iso-enzymes (1 mark)

...chromatography...

2. Certain factors have effects on proteins. For example, high temperature denatures most enzymes and it is for this reason that transportation of blood samples would normally be done at reduced temperatures.

i. Explain how denaturation of an enzyme affects its activity (1 mark)

Denaturation slows the reaction.

ii. Define optimum temperature for an enzyme (0.5 mark)

It is the temperature at which an enzyme works best.

iii. How does change in pH of the environment affect enzyme activity? (1 mark)

It can either favour or speed up an enzymatic reaction at optimum acidic pH or Alkaline pH. If pH is very low or too high the enzyme is denatured.

iv. How does increase in temperature lead to increase in enzyme activity? (1 mark)

It increases the kinetic energy of molecules.

v. Which protein structure is least affected by a denaturant? Explain (1 mark)

Primary structure.

vi. Define denaturation (1 mark)

It is a process that stops the enzyme activity because the temperature is too high or the acidity is too low or the alkalinity is too high.

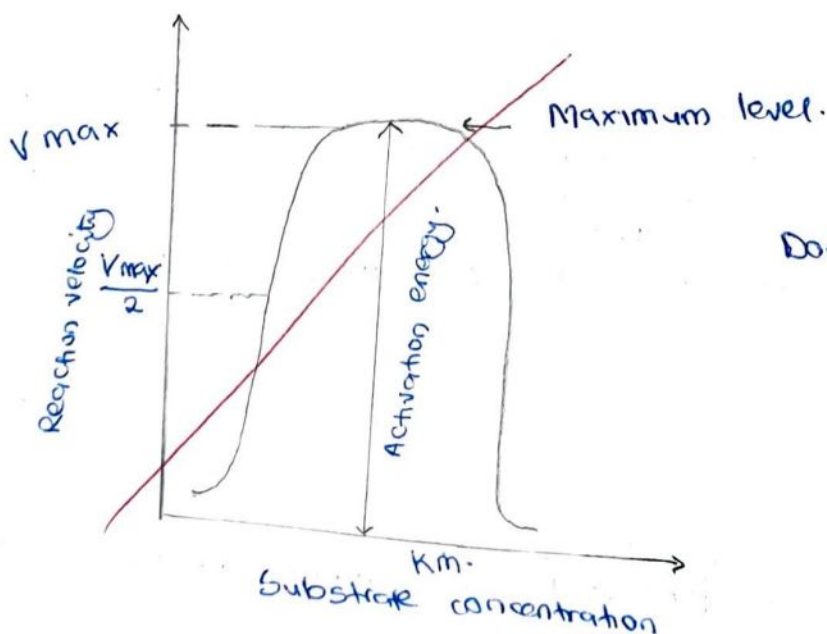
vii. List three properties exhibited by denatured proteins (1.5 marks)

Solubilization

Promote to enzymatic action

It cannot continue the reaction.

3. Draw a graph of reaction velocity (on the y-axis) of an enzyme catalysed reaction versus substrate concentration (x-axis) (1 mark)



Dome shaped reaction

- i. On the graph, indicate the following:  $K_m$ ,  $V_{max}$  and  $V_{max}/2$  (1 mark)
- ii. Explain  $V_{max}$  and  $K_m$  (1 mark)

$V_{max}$  is the maximum velocity of an enzymatic reaction.

$K_m$  is the substrate concentration in an enzymatic reaction at maximal velocity.

- iii. Explain the three segments of the graph (3 marks)

1. The beginning of the curve is  $V_0$  meaning the reaction has not yet begun.

2. Activation energy is the energy between  $V_0$  and  $V_{max}$ .  
Reaction velocity - The speed at which the reaction is taking place.

3.  $V_{max}$  is the equilibrium of the enzymatic reaction.

4. Substrate concentration is the amount of substrate in an enzymatic reaction.

- iv. Explain what happens when  $V_{max}$  is reached (1 mark)

When  $V_{max}$  is reached, the enzymatic reaction slows

down because all the enzyme active sites would have been bound and there is no enzyme remaining.

4. Proteins do everything in living cells. They are responsible for many different functions in a cell, making function one of the basis for classification. The building blocks of proteins are amino acids. Most of the amino acids in proteins are optically active.

- i. Based on composition, what are the classes of proteins? Define each class (2 marks)

Hydrophobic - They cannot dissolve in water (water insoluble)

Hydrophilic - They are water soluble

They are charged - Either negatively or positively charged

They can be zwitterion

ii. Based on the shape, what are the classifications of proteins? Define each class (2 marks)

Imidazole ring containing proteins  
Guanidino ring containing proteins

iii. Based on the structure, what are the classifications of proteins? Define each class (2 marks)

Primary Structure: This is the linear sequencing of amino acid residues in a polypeptide chain. Stabilized by Disulfide and Peptide bond.  
Secondary Structure: This is the folding of polypeptide chain along its long axis. Stabilized by hydrogen bonds.  
Tertiary: This is the 3 dimension folding of a protein.  
Quaternary: This has 2 or more polypeptide chains e.g. hemoglobin with 2 alpha polypeptide and 2 beta polypeptide.

iv. Name an amino acid that is optically active and is found in human protein (0.5 mark)

L - amino acid

v. Name two properties of an optically active amino acid (1 mark)

- They rotate Plane Polarized light to the left and right.
- They are non superimposable image of each other.

5. A mutation in a gene that codes for the  $\beta$  chain in hemoglobin causes sickle cell disease, a qualitative hemoglobinopathy. This results in interruption of oxygen flow to tissues. In hemoglobin and myoglobin there is a structure of complex protoporphyrin IX and  $\text{Fe}^{2+}$ .

i. Which amino acid is substituted in this change in sickle cell disease? (1 mark)

Glutamic acid is substituted by valine

ii. Explain how/why there is interruption of oxygen flow to tissues (2 marks)

The sickle celled red blood cells have a low surface area because of their shape hence the oxygen level is low.

- iii. Why is this disease classified as qualitative? (1 mark)  
 It is a matter of not following the amino acid sequence.
- iv. Name the product of oxidation of heme component of hemoglobin to give  $\text{Fe}^{3+}$  (0.5 mark)  
 methemoglobin
- v. In the oxidised state, is oxygen bound to the heme? (0.5 mark)  
 No
- vi. Why do you need the Iron in heme to be in a reduced state? (1 mark)  
 To be an electron carrier e.g. carrying of oxygen

6. One of the collagen diseases presented in class is Ehlers-Danlos syndrome. It is a heterogenous generalised connective tissue disorder with inheritable defects, stretchy skin and loose joints. Lack of hydroxylation of two amino acids is known to cause this syndrome.

- i. Name the two amino acids that are hydroxylated to stabilise the structure of collagen (2 marks)  
 lysine and proline
- ii. Which amino acid is glycosylated prior to formation of the triple helix in collagen? (1 mark)  
 Glycine
- iii. In addition to oxygen, the hydroxylation process requires a certain vitamin. Name the vitamin (0.5 mark)  
 Vitamin C
- iv. Name the condition that arises when there is lack of hydroxylation, no crosslinking and reduced tensile strength of collagen fibrils (1 mark)  
 Scurvy

7. A polypeptide chain has to attain a native conformation as soon as it comes out of the protein synthesizing machinery. The exact mechanisms involved in protein folding are not well elucidated, however, there are stages that have been categorised.

- i. List the four stages that occur in protein folding (2 marks)  
 Assembling of amino acids and sequencing them  
 Joining them either by peptide or disulfide bonds  
 Linking them up for folding  
 Cutting them in required polypeptides
- ii. Which structure determines the self-assembling process (first stage)? (1 mark)  
 The primary structure
- iii. Name two protein folding enzymes involved in the folding (1 mark)  
 Isomerase  
 chaperones
- iv. Which proteins aid protein folding by preventing formation of aggregates? (1 mark)  
 Chaperons
- v. How do these proteins prevent formation of aggregates? (1 mark)  
 They come in between the process of protein folding and prevents aggregate formation which slows protein folding.

8. Sulfonamide antibiotics are used in the treatment of bacterial infections. Bacteria synthesizes folic acid from p-aminobenzoic acid (PABA). Sulfonamides contain sulfanilamide, a structural analog of PABA. When used as chemotherapeutic agents, sulfonamides block the synthesis of folic acid in bacteria.

- i. What type of inhibition is exhibited in this case? (1 mark)  
..... competitive inhibition .....
- ii. Does this type of inhibition affect the maximum velocity ( $V_{max}$ ) of an enzyme catalysed reaction? (0.5 mark)  
..... yes - it reduces  $V_{max}$  .....
- iii. What is the effect on Michaelis-Menten ( $K_m$ ) constant in this case? (1 mark)  
..... The  $V_{max}$  is reduced and the  $K_m$  is increased .....
- iv. How is this constant related to the affinity of enzyme for the substrate? (0.5 mark)  
..... The lower the  $V_{max}$  the higher the affinity of the enzyme for substrate and vice versa .....
- v. Give an example of an irreversible type of inhibition of an enzyme (0.5 mark)  
.....

END